

Howard Andrés Martínez Meza

Senior Data Engineer · DataOps & Healthcare Data Platforms

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Data engineer with 7+ years building and operating healthcare data infrastructure at Abarca Health LLC. Focused on pipeline orchestration, platform reliability, and toolchain automation — not analytics consumption. Core stack: Apache Airflow 3.x, dbt, Snowflake, Kubernetes, T-SQL, and SSIS. PCAP-31-03 certified Python developer. Thrives at the intersection of data engineering and DevOps, constructing the systems that make data products possible.

WORK EXPERIENCE

Senior Data Engineer

Abarca Health LLC

June 2023 – Present

San Juan, PR

- ▶ Architected migration of 40+ production DAGs from Stonebranch to Apache Airflow 3.x TaskFlow API, eliminating legacy scheduler dependency and improving end-to-end pipeline observability
- ▶ Engineered containerized dbt transformation runs via `@task.kubernetes_cmd`, enabling isolated, reproducible Kubernetes pod execution per transformation step
- ▶ Built dynamic DAG factory pattern for pharmaceutical data synchronization, cutting new-client pipeline onboarding from days to hours
- ▶ Led Blue Shield of California (BSC) platform deployments: ETL design, Snowflake schema validation, and cross-functional alignment from raw ingestion through reporting layer
- ▶ Onboarded LucyRx data pipelines, coordinating data contracts, transformation logic, and quality validation across engineering and product teams
- ▶ Designed Snowflake stored procedures with Time Travel recovery logic for audit-safe healthcare data rollbacks
- ▶ Earned PCAP-31-03 certification (December 2025); applied advanced Python patterns to standardize DataOps tooling and pipeline development across the team

Business Intelligence Specialist III — Team Lead

Abarca Health LLC

April 2022 – June 2023

San Juan, PR

- ▶ Led a 4-person BI team through the active migration from SSIS to dbt as the transformation layer, driving adoption of version-controlled, modular SQL models across the squad
- ▶ Championed the transition from SQL Server to Snowflake as the primary data platform; designed initial star schema models in Snowflake, reducing core PBM dashboard query time by 40%
- ▶ Maintained T-SQL and SSIS workloads in parallel during the transition period, ensuring zero disruption to client reporting SLAs while the new stack was validated
- ▶ Established dbt coding standards, testing patterns, and code review practices that became the team baseline for all new pipeline development

Business Intelligence Specialist II — Team Lead

Abarca Health LLC

April 2020 – April 2022

San Juan, PR

- ▶ Developed T-SQL stored procedures and SSIS pipelines ingesting pharmaceutical claims data from multiple payers, processing 1M+ records per daily cycle
- ▶ Initiated early evaluation of dbt and Snowflake as modern replacements for the SSIS/SQL Server stack; built proof-of-concept models that informed the platform migration roadmap
- ▶ Promoted to team lead; mentored 2 analysts on T-SQL optimization, SSIS package design, and data modeling fundamentals as the team prepared for the upcoming stack transition

Business Intelligence Specialist I

Abarca Health LLC

April 2019 – April 2020

San Juan, PR

- Built T-SQL queries, stored procedures, and SSIS ETL packages for the pharmacy analytics platform, supporting monthly reporting cycles for 5+ client accounts
- Automated recurring manual reporting workflows via SSIS, saving approximately 12 analyst hours per week

Junior Business Intelligence Specialist

Abarca Health LLC

April 2018 – April 2019

San Juan, PR

- Designed and implemented the company's first Master Data Services (MDS) solution, addressing a critical gap in reference data governance with no prior infrastructure in place
- Developed T-SQL scripts and SSIS packages for pharmacy benefit data extraction and transformation, supporting the go-live of two enterprise health plan clients

Professional in Systems

Coosalud EPS-S

November 2014 – July 2015

Cartagena, COL

- Developed internal database systems and technical solutions supporting health insurance operations for a national Colombian health provider

Young Researcher

University of Cartagena

May 2013 – May 2014

Cartagena, COL

- Conducted research on cloud computing and big data architectures; co-authored published IEEE conference paper on smart grid brokerage systems

National Assistant in Systems

Coosalud EPS-S

September 2012 – March 2013

Cartagena, COL

- Provided systems and technical support across national health insurance operations

TECHNICAL SKILLS

Orchestration: Apache Airflow 3.x (TaskFlow API, KubernetesPodOperator), Stonebranch

Pipeline / ETL: SSIS, dbt (models, tests, snapshots, macros), T-SQL stored procedures

Data Platform: Snowflake (Time Travel, Streams, Dynamic SQL, RBAC, Stored Procedures), SQL Server

Infrastructure: Kubernetes, Docker, Bash

Languages: Python (PCAP-31-03), T-SQL, SQL, Bash

Domain: Pharmacy benefit management (PBM), pharmaceutical analytics, healthcare data platforms

EDUCATION

Master of Business Intelligence

Online Business School & Universitat de Barcelona

July 2023

Spain

Master of Engineering in Computer Engineering

University of Puerto Rico, Mayaguez

GPA: 4.00 / 4.00

January 2018

Puerto Rico, PR

Systems Engineer

University of Cartagena

May 2014

Cartagena, COL

CERTIFICATIONS

PCAP-31-03 · Python Institute — Certified Associate in Python Programming · December 2025

PUBLICATIONS

Cloud-based and Big Data-enabled Brokerage System for Smart Grids — IEEE BigData Congress 2017 (6th International Congress on Big Data)

LANGUAGES

Spanish (Native) **English** (Fluent)